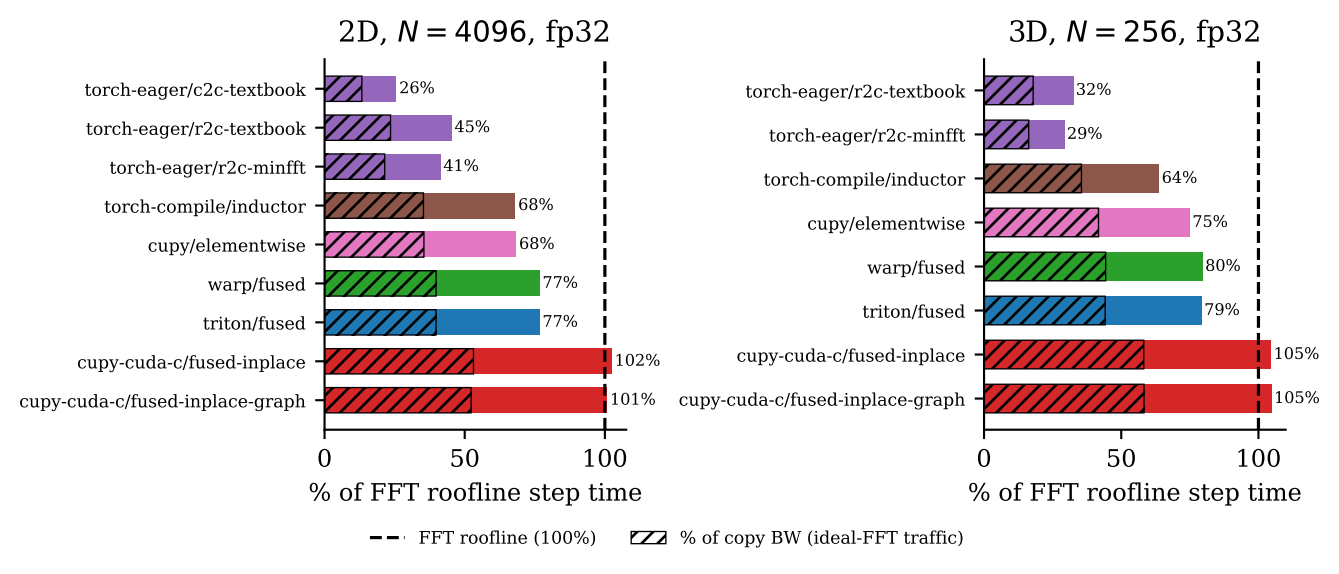
2D, $N = 4096$, fp32
torch-eager/c2c-textbook
torch-eager/r2c-textbook
torch-eager/r2c-minfft
torch-compile/inductor
cupy/elementwise
warp/fused
triton/fused
cupy-cuda-c/fused-inplace
cupy-cuda-c/fused-inplace-graph
26%
45%
41%
68%
68%
77%
77%
102%
101%
0
50
100
% of FFT roofline step time
3D, $N = 256$, fp32
torch-eager/r2c-textbook
torch-eager/r2c-minfft
torch-compile/inductor
cupy/elementwise
warp/fused
triton/fused
cupy-cuda-c/fused-inplace
cupy-cuda-c/fused-inplace-graph
32%
29%
64%
75%
80%
79%
105%
105%
0
50
100
% of FFT roofline step time
FFT roofline (100%)
% of copy BW (ideal-FFT traffic)